CORE CI/CD + SECURITY)

1️⃣ actions/checkout

```
- uses: actions/checkout@v4
```

💡 Clones your repository into the runner
🔧 Used in: EVERY pipeline
🎯 GH-200: First step in workflow execution

2️⃣ actions/setup-node

```
- uses: actions/setup-node@v4
  with:
    node-version: 18
```

💡 Installs Node.js runtime
🔧 Used in: Node.js apps
🎯 GH-200: Language setup actions

3️⃣ actions/setup-java

```
- uses: actions/setup-java@v4
  with:
    distribution: temurin
    java-version: 17
```

💡 Prepares Java environment
🔧 Used in: Maven / Gradle builds

4️⃣ actions/setup-python

```
- uses: actions/setup-python@v5
  with:
    python-version: '3.10'
```

💡 Installs Python runtime
🔧 Used in: backend / ML pipelines

5️⃣ actions/cache

```
- uses: actions/cache@v4
  with:
    path: ~/.npm
    key: cache-${{ hashFiles('package-lock.json') }}
```

💡 Stores dependencies to speed builds
🔧 Used in: performance optimization
🎯 GH-200: Cache vs artifact

6️⃣ actions/upload-artifact

```
- uses: actions/upload-artifact@v4
  with:
    name: build
    path: dist/
```

💡 Saves build output
🔧 Used in: multi-job pipelines

7️⃣ actions/download-artifact

```
- uses: actions/download-artifact@v4
  with:
    name: build
```

💡 Retrieves stored files
🔧 Used in: cross-job communication

8️⃣ actions/github-script

```
- uses: actions/github-script@v7
  with:
    script: console.log("Hello")
```

💡 Run JavaScript inside workflow
🔧 Used in: automation logic

9️⃣ docker/login-action

```
- uses: docker/login-action@v3
  with:
    username: ${{ secrets.DOCKER_USERNAME }}
    password: ${{ secrets.DOCKER_PASSWORD }}
```

💡 Login to Docker registry
🔧 Required before pushing images

🔟 docker/build-push-action

```
- uses: docker/build-push-action@v5
  with:
    push: true
    tags: myapp:latest
```

💡 Build & push Docker images
🔧 Core DevOps pipeline step

1️⃣1️⃣ aws-actions/configure-aws-credentials

```
- uses: aws-actions/configure-aws-credentials@v4
  with:
    aws-region: us-east-1
```

💡 Authenticate to AWS
🔧 Used before deployments

1️⃣2️⃣ hashicorp/setup-terraform

```
- uses: hashicorp/setup-terraform@v3
```

💡 Install Terraform CLI
🔧 Used in infra pipelines

🔐 SECURITY (VERY IMPORTANT)

1️⃣3️⃣ github/codeql-action/init

```
- uses: github/codeql-action/init@v3
  with:
    languages: javascript
```

💡 Initializes security scanning
🔧 Used in secure CI pipelines
🎯 GH-200: Security concept

1️⃣4️⃣ github/codeql-action/analyze

```
- uses: github/codeql-action/analyze@v3
```

💡 Runs vulnerability scan
🔧 Detects code issues

1️⃣5️⃣ aquasecurity/trivy-action

```
- uses: aquasecurity/trivy-action@v0.20.0
```

💡 Scans Docker images for vulnerabilities
🔧 Used in container pipelines

🐳 DOCKER ADVANCED

1️⃣6️⃣ docker/setup-buildx-action

```
- uses: docker/setup-buildx-action@v3
```

💡 Enables advanced Docker builds
🔧 Multi-platform images

1️⃣7️⃣ docker/metadata-action

```
- uses: docker/metadata-action@v5
```

💡 Auto-generates image tags
🔧 Versioning strategy

☁️ CLOUD DEPLOYMENT

1️⃣8️⃣ aws-actions/amazon-ecr-login

- uses: aws-actions/amazon-ecr-login@v2

💡 Login to AWS container registry
🔧 Required before push

1️⃣9️⃣ aws-actions/ecs-deploy-task-definition

- uses: aws-actions/ecs-deploy-task-definition@v2

💡 Deploy containers to ECS
🔧 Production deployment

2️⃣0️⃣ google-github-actions/deploy-cloudrun

- uses: google-github-actions/deploy-cloudrun@v2

💡 Deploy to Cloud Run
🔧 Serverless deployment

🧪 QUALITY + TESTING

2️⃣1️⃣ sonarsource/sonarqube-scan-action

- uses: sonarsource/sonarqube-scan-action@v2

💡 Code quality analysis
🔧 Used in quality gates

2️⃣2️⃣ codecov/codecov-action

- uses: codecov/codecov-action@v4

💡 Upload test coverage
🔧 CI reporting

⚙️ EXTRA REAL-WORLD (IMPORTANT)

2️⃣ 3️⃣ azure/login

- uses: azure/login@v2

💡 Login to Azure
🔧 Multi-cloud pipelines

2️⃣ 4️⃣ google-github-actions/auth

- uses: google-github-actions/auth@v2

💡 Authenticate to GCP
🔧 Cloud pipelines

2️⃣ 5️⃣ styfle/cancel-workflow-action

- uses: styfle/cancel-workflow-action@v0.12.0

💡 Cancels duplicate runs
🔧 Cost optimization